

Day 1 — The Moon and Its Shapes

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: observe and name the four basic phases (new, first quarter, full, last quarter).

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *candra* (IAST: candra; lit. “Moon”)

Primary source (today’s reading)

From *Sūrya-siddhānta* (c. 5th–10th c. CE) — see sources.md.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Geometric model.** Use a torch (Sun) + tennis ball (Moon) + a student’s head (Earth). Walk the Moon around the Earth. “*Which part of the Moon does the Earth-observer see lit at each position?*” Demonstrate the four primary phases.
2. **Define synodic month.** ~29.53 days from one *amāvasyā* to the next. *Synodic* = relative to the Sun (as seen from Earth). Distinct from *sidereal month* (~27.32 days) — relative to fixed stars.
3. **Source.** *Sūrya-siddhānta*, ch. 2, vv. 1–10 — defines lunar motion with mean and true longitudes. Read 2 verses aloud in transliteration.
4. **Four phases:**
 - *amāvasyā* — new (Sun side)
 - First quarter — half lit, waxing
 - *pūrṇimā* — full (anti-Sun side)

- Last quarter — half lit, waning
- 5. **Class observation assignment.** Each student notes the moon’s shape this evening (or tomorrow morning) and writes one sentence describing it. We’ll plot data tomorrow.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces The Moon and Its Shapes.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).